



LONGEVITY & ANTI-AGING

GHK-Cu

GHK-Cu – 100 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Glycyl-L-histidyl-L-lysine - copper(II) complex
SYNONYMS / ALIASES	GHK-Cu; copper tripeptide-1; copper peptide GHK-Cu2+
CAS NUMBER	89030-95-5 (copper complex); 49557-75-7 (GHK peptide free acid)
MOLECULAR FORMULA	C14H22N6O4 with bound Cu (1:1 complex)
MOLECULAR WEIGHT	Approximately 401.92 g/mol (Cu complex); 340.38 g/mol (free peptide)
AMINO ACID SEQUENCE	GHK (3 residues; H-Gly-His-Lys-OH) coordinated to a Cu(II) ion
SOURCE	Synthetic (solid-phase peptide synthesis followed by Cu(II) complexation)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized blue powder (color from the Cu(II)-tripeptide complex)
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA); copper content per ICP analysis
CONTAINER	100 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	2-8 C or -20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only); avoid strongly acidic solvents which may displace Cu(II)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months
STABILITY NOTES	The Cu(II) coordination is pH-dependent and can dissociate at low pH or in the presence of strong chelators (EDTA, citrate at high concentration). Solution color (blue) is a visual indicator of complex integrity.

STRUCTURAL NOTES

GHK-Cu is a 1:1 coordination complex between the tripeptide glycyl-L-histidyl-L-lysine (GHK) and a divalent copper ion (Cu(II)). The GHK tripeptide is a naturally occurring sequence found in human plasma, saliva, and urine. In the complex, Cu(II) is coordinated by three GHK donor atoms (the imidazole nitrogen of histidine, the alpha-amine of glycine, and the deprotonated amide nitrogen of the Gly-His peptide bond), with a fourth coordination site available for water or other ligands.

SELECTED PEER-REVIEWED REFERENCES

1. Pickart L., Margolina A. - *Regenerative and protective actions of the GHK-Cu peptide in the light of the new gene data* - International Journal of Molecular Sciences, 2018 - DOI: 10.3390/ijms19071987
2. Pickart L. et al. - *GHK peptide as a natural modulator of multiple cellular pathways in skin regeneration* - BioMed Research International, 2015 - DOI: 10.1155/2015/648108
3. Maquart F.X. et al. - *Stimulation of collagen synthesis in fibroblast cultures by the tripeptide-copper complex glycyl-L-histidyl-L-lysine-Cu(II)* - FEBS Letters, 1988 - DOI: 10.1016/0014-5793(88)80445-5

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

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Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.